Project Progress Report

Unsupervised Learning Project | Milestone 2

Team	**Francisco Rodrigues dos Santos (72908) · Duarte Ferreira (75419)**
Project title	Hotel Booking Demand: Clustering of Booking Behaviour Profiles
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Repository	*[GitHub URL — TBD]*
Repository version shown in lab	*[commit hash / tag — TBD]*

1. RQ1 and segmentation time

Segmentation question

What recurring booking-behaviour profiles — defined over stay length, seasonality, distribution channel and commitment level — emerge from hotel reservation records, using only information available at the moment of booking confirmation? The goal is an exploratory segmentation usable for operational and marketing decisions at booking time. Clustering is justified because no supervised label captures "behaviour at booking time".

Unit of analysis

One row equals one booking record (not a customer). Conclusions are framed as recurring booking behaviours, not customer types.

Segmentation (index) time

Booking confirmation. Every clustering input is available at or before that point. Anything updated afterwards (cancellation outcome, room re-assignment, waiting-list days, post-event status) is reserved for post-hoc profiling only.

2. Dataset version control and documentation

Course release v1 (`hotel_bookings_course_release_v1.csv`), 119,390 rows by 32 columns. SHA-256 fingerprint `7c2ae42a...c7b1fc06`, recorded in `data/raw/DATASET_MANIFEST.yml`. Source: Kaggle (`jessemostipak/hotel-booking-demand`, CC BY 4.0); peer-reviewed reference António, de Almeida and Nunes (2019), Data in Brief, doi:10.1016/j.dib.2018.11.126. Raw CSV is not committed.

Known data-quality issues

- `company` missing in 94.3 % of rows and `agent` in 13.7 %; both are ID-like and dropped.
- `country` missing in 0.41 % and high-cardinality (~93 codes); rare-grouped at minimum frequency 100.
- `meal == "Undefined"` and a few negative `adr` values are recoded to NaN before imputation.
- Heavy right-skew in `lead_time`, `previous_cancellations`, `previous_bookings_not_canceled`, which is why we keep RobustScaler as a co-equal scaler variant.
- The four raw `arrival_date_*` columns are flagged by the course's `column_roles.csv` for cyclic encoding.

3. Completed Exploratory Data Analysis (EDA)

Missingness summary

Column	Type	n_missing	%
`company`	categorical	112,593	94.31
`agent`	numerical (ID)	16,340	13.69
`country`	categorical	488	0.41
`children`	numerical	4	0.003
All other 28 columns	—	0	0.00

Outlier summary (1.5·IQR fences, numerical clustering inputs)

`lead_time` (Q1=18, Q3=164, fence 383), `stays_in_week_nights` (Q1=1, Q3=3, fence 6) and `stays_in_weekend_nights` (Q1=0, Q3=2, fence 5) are kept: long lead-times and long stays are real booking behaviour. The remaining numerical inputs (`adults`, `children`, `babies`, `previous_cancellations`, `previous_bookings_not_canceled`) all have zero IQR — non-zero values are exactly the signal we want, so rows are not removed. Heavy tails are absorbed by median imputation plus Standard or Robust scaling.

Variables excluded from clustering inputs

- **Mandatory leakage exclusions** (post-event): `is_canceled`, `reservation_status`, `reservation_status_date`.
- **Post-confirmation exclusions**: `assigned_room_type`, `booking_changes`, `days_in_waiting_list`.
- **Identifier exclusions**: `agent`, `company`.
- **Re-coded into cyclic pairs and then dropped from the clustering input**: `arrival_date_month` becomes (`arrival_month_sin`, `arrival_month_cos`); `arrival_date_week_number` becomes (`arrival_week_sin`, `arrival_week_cos`).
- **Dropped outright**: `arrival_date_year` (3-value dataset-window artefact, Jul-2015 to Aug-2017) and `arrival_date_day_of_month` (sub-monthly noise).
- **Profiling-only** (kept in the profiling frame, never enter the distance computation): `meal`, `required_car_parking_spaces`, `total_of_special_requests`, `adr`. Price is a downstream consequence of the booking choices we cluster on; mixing it in conflates cluster *definition* with cluster *description*.

All exclusions are centralised in `src/preprocessing/feature_config.py`.

4. Main representation (RQ2)

Included variables

- **Numerical (13)**: `lead_time`, `stays_in_weekend_nights`, `stays_in_week_nights`, `adults`, `children`, `babies`, `is_repeated_guest`, `previous_cancellations`, `previous_bookings_not_canceled`, plus the four cyclic features `arrival_month_sin`, `arrival_month_cos`, `arrival_week_sin`, `arrival_week_cos`. `is_repeated_guest` (binary 0/1) stays in the numerical block as a single column rather than being one-hot-encoded into two collinear columns.
- **Categorical (7)**: `hotel`, `country`, `market_segment`, `distribution_channel`, `reserved_room_type`, `deposit_type`, `customer_type`.

Excluded and profiling-only variables

Leakage and identifier exclusions are listed in section 3 (Milestone2_Report.md:56-58). Specific to the representation, four variables are kept in the profiling frame but never enter the distance: `meal`, `required_car_parking_spaces`, `total_of_special_requests`, `adr`. Each is observable at booking time, so it is not a leakage case; the reason for exclusion is that they are consequences of the booking choices we cluster on (price, on-site requests). Including them would conflate cluster definition with cluster description and let the descriptors do the modelling work.

Numerical preprocessing

Median imputation, then scaling. Cyclic encoding for both month and week-of-year so December and January (or week 52 and week 1) are close in feature space rather than 11 or 51 units apart. Two scaler variants run side-by-side under the same protocol: **StandardScaler** (zero mean, unit variance) and **RobustScaler** (median, IQR-based, resistant to long tails). We report both so any conclusion has to hold up under either.

Categorical preprocessing

Imputation with constant value `"Unknown"`. A custom `RareCategoryGrouper` collapses categories below 50 occurrences (100 for `country`) into `"Other"` before encoding. `OneHotEncoder` then maps unseen categories at scoring time to all-zeros, in a single encoding pass without double-counting.

Final representation

`representation_id = repA`. Implied geometry is **Euclidean** on a dense matrix of 13 numerical plus about 40 one-hot dimensions, totalling 53 features on the 5,000-row fast-check. Internal indices in section 5 are computed in this same space, with no metric switch in between.

5. Baseline modelling evidence

Methods and configuration

- **Method 1**: MiniBatch K-Means (`n_init=10`, `batch_size=1024`, `max_iter=300`). The centroid update rule and Euclidean objective are identical to full-batch K-Means; only the per-step subsampling differs. Mini-batch is needed at n = 119k for a tractable seed sweep.
- **Method 2**: iK-means (Mirkin, 2005); auto-determines K via anomalous-pattern peeling on a PCA-projected discovery space, then refits in the full 53-dim space.
- **Predefined K range**: {2, 3, 4, 5, 6, 7, 8}; bounded a-priori (below 2 trivial, above 8 not interpretable as an RQ1-actionable segmentation for operational and marketing decisions at booking time).
- **Seeds**: ten lock-stepped seeds {0..9}, giving 45 ARI pairs per (method, scaler, K) cell.
- **Distance**: Euclidean. **Silhouette** is sub-sampled (2,000 / 5,000 points per seed) to keep its $O(n^2)$ cost tractable.

Internal indices (5,000-row fast-check, p = 53, mean ± std over 10 seeds)

K	Sil. (Std)	Sil. (Rob)	DB (Std)	DB (Rob)	CH (Std)	CH (Rob)
2	**0.1332 ± 0.0041**	0.0974 ± 0.0214	2.59	3.08	581	347
3	0.1229 ± 0.0059	0.0878 ± 0.0299	2.24	2.63	519	367
4	0.1210 ± 0.0082	0.0884 ± 0.0167	2.12	2.46	475	397
5	0.1200 ± 0.0142	0.0864 ± 0.0177	2.00	2.32	459	421
6	0.1130 ± 0.0199	0.0775 ± 0.0192	2.00	2.31	446	388
7	0.1140 ± 0.0160	0.0798 ± 0.0134	1.85	2.21	450	373
8	0.1101 ± 0.0185	0.0761 ± 0.0122	1.85	2.20	442	375

DB and CH cells show the seed-mean only; per-seed std is logged in `experiments.csv` and is omitted here for table compactness.

K = 2 maximises silhouette under both scalers. CH and DB favour higher K (as expected; they reward smaller, denser clusters), but their gain plateaus from K = 5. The cyclic-week feature added at this checkpoint tightened the silhouette spread substantially: standard-scaler std at K = 2 dropped from 0.033 (5 seeds, no week cyclic) to 0.004 (10 seeds, with week cyclic) — about 8x less seed-to-seed noise on the headline metric.

iK-means (auto-K) and runtime

iK-means picked K = 4 under StandardScaler (silhouette 0.144) and K = 3 under RobustScaler (silhouette 0.847). The very high robust-scaler silhouette is plausibly an artefact of anomalous-pattern peeling on long-tailed inputs (heavy outliers form tight far-away mini-clusters) and is flagged for follow-up under the full pass.

The fast-check pipeline (`bash run_all --fast`, Windows 11, Python 3.13) runs k-means and iK-means end-to-end in about 2 minutes (140 k-means fits in ~90 s, 20 iK-means fits in ~15 s). The full 119,390-row pass is part of the 13-May Core-Protocol-Freeze.

First cluster-interpretation note (StandardScaler, K = 2)

The two clusters split along stay-length, seasonality and deposit:

- **Cluster 0 — short-stay, online-distributed, low-commitment**: lead-time about 70 days median, total nights about 2.9, online-TA share 59 %, no-deposit share 93 %, non-refundable 7 %.
- **Cluster 1 — long-stay, offline / groups, higher non-refundable**: lead-time about 130 days median, total nights about 7.7, online-TA share 28 %, no-deposit share 78 %, non-refundable 21 %.

Both clusters have non-trivial sizes, and `is_canceled` (held out from clustering) varies materially across them in the post-hoc profiling layer. That is a first sign the partition is picking up real behavioural variance and not just noise.

6. Sensitivity / robustness plan

Main representation: `repA` (defined in section 4). The variants below are perturbations of `repA`.

Resampling rule: 10 lock-stepped seeds `{0..9}` per (method, scaler, K) cell, giving the 45 ARI pairs reported below. Same seed scheme as section 5.

Stability evidence so far

Stability is measured as **mean pairwise Adjusted Rand Index** over 45 seed pairs at fixed (method, scaler, K). Selection rule: ARI of at least 0.80 to accept a K as stable.

K	ARI (StandardScaler)	ARI (RobustScaler)
2	0.334	0.171
5	**0.468**	0.401
7	0.422	**0.436**

No K reaches the 0.80 threshold yet — the partition is silhouette-informative but seed-fragile. This is the central robustness question that motivates the planned variants below.

Controlled variants planned

Three comparisons, anchored against `repA`-StandardScaler:

1. `repA`-standard vs `repA`-robust — already running in the same pipeline; the table above quantifies it. Does the silhouette-best K and its stability survive a scaler swap from Standard to Robust?
2. `repA` vs an engineered representation `repB` (in `pipeline_v2.py`: `total_nights`, `party_size`, `has_kids`, `weekend_share`, plus `VarianceThreshold` on rare dummies and block-equalised numerical-vs-categorical contribution). Does the K = 2 silhouette dominance survive feature engineering that re-balances the implicit Euclidean metric?
3. `repA` with vs without PCA-20. Does compressing the 53-dim space to 20 PCs change which K is silhouette-best? Particularly relevant for iK-means, whose discovery phase already runs in PCA space.

Acceptance bar. A partition is considered "robust" only if it (i) maximises silhouette in at least 3 of the 4 variants and (ii) reaches mean ARI of at least 0.80 in at least one variant.

7. Experiment logging and repository status

`experiments.csv` is auto-appended at every run via `src/utils/experiment_logger.py` (300 rows after the latest run). Schema: `task, method, variant, k, seed, silhouette, calinski_harabasz, davies_bouldin, ari_vs_seed0, bic, aic, log_likelihood, n_iter, converged, notes`. A `date / run_id` pair, an explicit `representation_id` column, a `parameters` JSON column (full hyperparameter vector), and a `sample_rule` column (`fast-check-5k` vs `full-119k`) will be added before the 13-May Core-Protocol-Freeze.

`environment.yml` (conda, Python 3.11) is checked in; a pinned `requirements.txt` for the actual lab environment (Python 3.13 on Windows 11) is being prepared from `pip freeze` for the 20-May Reproducibility Freeze.

`run_all` (bash) and **`run_all.ps1`** (PowerShell wrapper for native Windows) execute the Task 1 pipeline end-to-end: SHA-256 validation, preprocessing, baseline clustering with k-means and iK-means under both scalers. Every figure in `figures/` and every table in `tables/` is regenerated without manual steps.